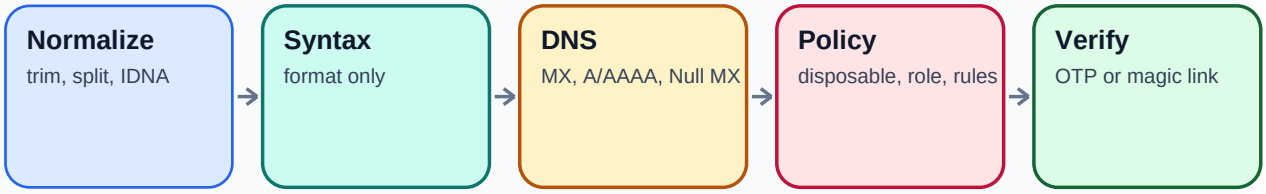
Email Checker
Wide Research Report

How email validation really works, what it can and cannot prove, and what to build first if you want a practical production system.

What a modern email checker actually does



Synthesis: syntax stops obvious garbage; OTP is what proves inbox control.

Bottom line: the most reliable architecture is layered. Syntax and HTML input validation catch formatting issues. DNS checks establish whether the domain can likely receive mail. Policy checks catch disposable or risky domains. OTP or magic-link verification is what actually proves the user controls the inbox. SMTP probing can add signal, but it should be treated as optional and probabilistic rather than as the foundation of the product. [S1][S2][S3][S5]

Prepared	April 2, 2026
Scope	Protocol standards, DNS and SMTP behavior, product design, deliverability, build strategy
Output	Wide research PDF with original diagrams and implementation guidance

Research note

This report uses official standards and primary documentation where possible. Places where the report goes beyond the standards into implementation strategy are explicitly presented as analyst synthesis rather than protocol fact.

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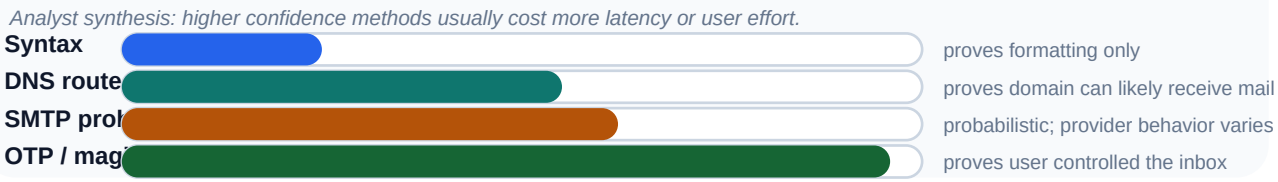
Executive Summary

- An email checker is not one thing. It is a stack of checks that answer different questions at different confidence levels.
- RFC 5322 defines the core email shape, while RFC 5321 governs how SMTP systems treat local parts, domains, delivery routing, and recipient handling. [S1][S2]
- DNS is the best technical midpoint for an MVP. MX lookup, A or AAAA fallback, and Null MX detection give strong routing signal without SMTP latency or anti-abuse friction. [S2][S3][S6]
- SMTP probing is useful only as a secondary signal because servers can refuse VRFY, defer decisions, accept mail for later rejection, or behave like catch-all domains. [S2]
- If the product goal is to confirm the user owns the inbox, OTP or magic-link verification is the decisive control. The rest of the checker only narrows risk. [S2][S9]
- Teams that send verification emails must think beyond validation. Deliverability now depends on SPF or DKIM for all Gmail-bound senders, and SPF plus DKIM plus DMARC for high-volume Gmail senders, with valid DNS and TLS requirements called out by Google. [S9]

Method	What it proves	Strength	Primary weakness
HTML / regex	Looks like an email	Fastest UX filter	No deliverability proof
Syntax library	Server-side format check	Reliable baseline guardrail	Still no mailbox proof
DNS route check	Domain likely accepts mail	Best low-friction protocol signal	Cannot prove inbox ownership
SMTP probe	Mailbox may exist	Sometimes catches dead inboxes	Provider behavior is inconsistent
OTP / magic link	User controlled the inbox	Highest confidence	Adds user friction

Evidence strength by validation method



Decision framing

There are really three different products people call an email checker: a formatter, a deliverability pre-checker, and an ownership verifier. Conflating them leads to incorrect promises in the UI and unnecessary engineering complexity.

Standards and Technical Deep Dive

1. Syntax and normalization

RFC 5322 defines the core address form as `local-part@domain`. The local part may be more permissive than many developers expect, and RFC 5321 adds an important operational detail: mailbox local parts must be preserved as case-sensitive even though many providers are case-insensitive in practice. [S1][S2]

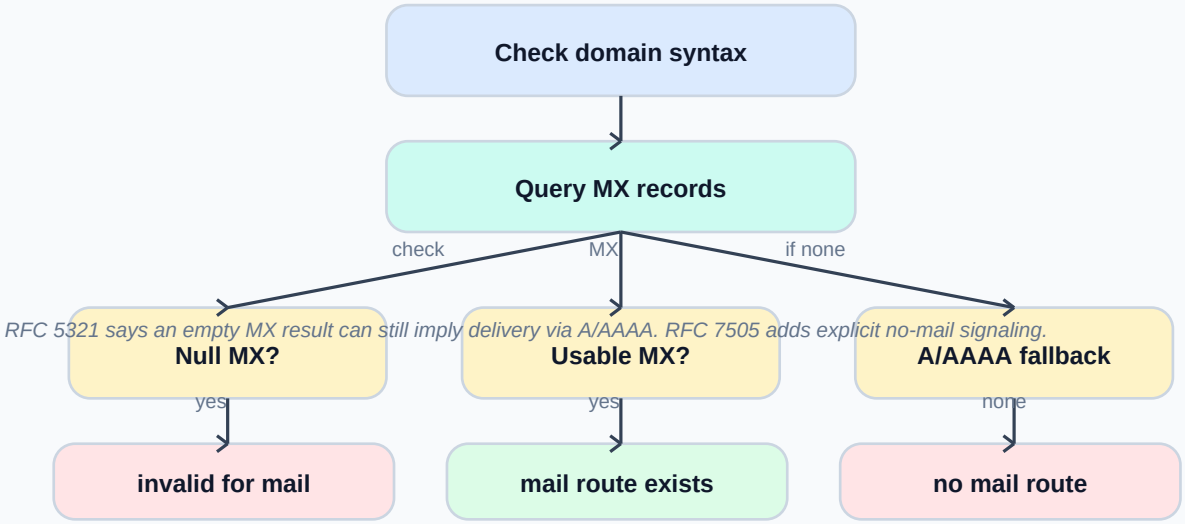
This is why practical implementations should avoid home-grown mega-regexes and should be careful with canonicalization. Browser validation via `input type=email` is useful for fast UX, but MDN is clear that it only checks formatting. On the server side, a library such as validator.js provides a safer baseline and exposes normalization options that reflect real provider behavior. The validator.js documentation also warns that lowercasing every local part may violate RFC 5321. [S5][S7]

2. DNS routing matters more than most teams think

RFC 5321 specifies the core MX lookup behavior. A domain with usable MX records can likely receive mail through those targets. A less obvious rule is the implicit MX fallback: if an MX lookup returns an empty list, SMTP clients are supposed to treat the address as if it had an implicit MX pointing back to the host itself, then rely on A or AAAA records. That means no MX is not always equivalent to no mail route. [S2]

RFC 7505 refines this story with Null MX. A domain that publishes a single MX record with preference 0 and exchange "." is explicitly saying that it does not accept mail. That is a much stronger signal than simply not finding MX records. [S3]

DNS mail-route decision flow



3. Internationalized mailboxes

If you need to support non-ASCII mailbox names or internationalized domains beyond basic IDNA handling, RFC 6531 introduces SMTPUTF8. This matters because some teams assume Unicode email is just a UI problem, when in reality it can change downstream SMTP compatibility and vendor choices. A pragmatic MVP may start with ASCII local parts and IDNA-converted domains, then add SMTPUTF8 if the market requires it. [S4]

4. SMTP probing is real, but not decisive

SMTP probing typically uses EHLO, MAIL FROM, and RCPT TO against the domain's chosen mail exchanger. In theory it answers a compelling question: does this mailbox exist? In practice the protocol and modern provider behavior make the answer unreliable. RFC 5321 defines the 252 response as a case where the server will accept mail but will not confirm the recipient in advance, and it allows sites to disable VRFY or EXPN for security reasons rather than reveal valid recipients. [S2]

SMTP reality	What it means for your checker
252 Cannot VRFY user	The server will accept mail but will not confirm the recipient in advance.
VRFY or EXPN disabled	Many servers deliberately avoid easy recipient enumeration for security reasons.
Catch-all domains	RCPT acceptance can look valid even when the mailbox may not exist as an individual account.
Delayed rejection	Some systems decide only after message body or internal policy checks, so signup-time probing can mislea

Analyst synthesis

SMTP probing should be treated as a background signal, not as the product's promise to the user. The checker can say deliverable likely or high risk more honestly than it can say mailbox definitely exists.

5. Disposable domains and policy layers

Temporary or throwaway mail detection is not part of SMTP or RFC syntax. It is a policy choice. Public projects such as disposable-email-domains maintain blocklists specifically for domains that are often used for spam, dummy registrations, or abuse, and the repo notes real production use cases such as PyPI account protection. [S8]

The same policy layer is where role accounts such as `admin@`, `info@`, or `support@` belong. They are not invalid by protocol, but they may be lower-quality leads or higher-risk registrations depending on the product.

6. Deliverability of your own verification email

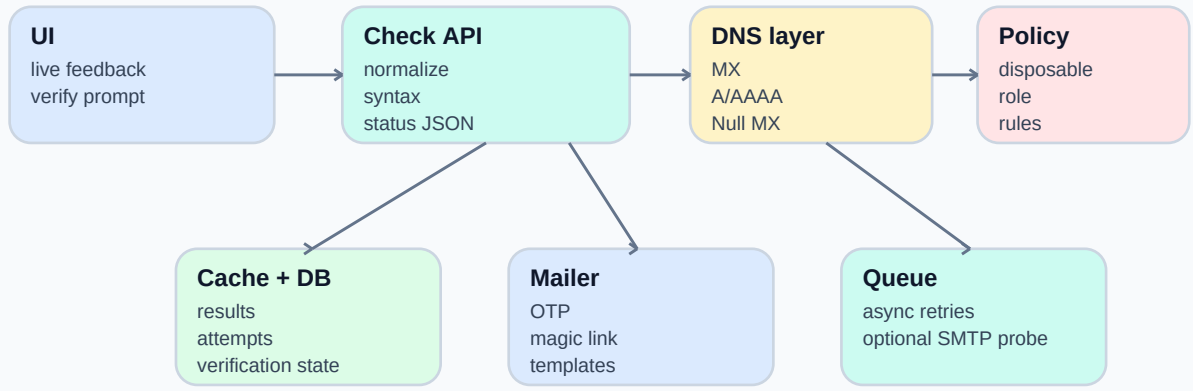
Once you choose OTP or magic-link verification, your checker becomes a sender. At that point the engineering problem expands from validation to sender reputation, DNS hygiene, and authentication. Google's current sender guidance is especially important because verification and onboarding flows frequently target Gmail inboxes. [S9]

Gmail requirement snapshot	Current official guidance
All senders	Starting February 1, 2024, Gmail requires SPF or DKIM, valid forward and reverse DNS, TLS, RFC 5322
5,000+ messages per day	Starting February 1, 2024, Gmail additionally requires SPF and DKIM, DMARC, alignment, and one-click
Operational implication	An email checker that sends OTPs at scale must be paired with solid sender-domain operations, not just

Implementation Guidance

For most products, the strongest first milestone is a layered MVP with these responsibilities: client formatting feedback, server-side syntax validation, DNS mail-route checks, disposable-domain and role-account policy checks, and OTP or magic-link verification. Keep SMTP probes out of the signup critical path unless the business case is unusually strong.

Recommended service architecture



Design principle: keep signup latency on syntax + DNS, and move slower checks to async workflows.

Recommended API shape

A useful API should return structured states rather than a single boolean. This makes the front end more honest and gives you room to add new checks without changing the contract. Example statuses include `invalid_syntax`, `null_mx`, `disposable_domain`, `deliverable_likely`, and `verified`.

```
{
  "normalizedEmail": "user@example.com",
  "status": "deliverable_likely",
  "signals": {
    "syntaxValid": true,
    "domainResolvable": true,
    "mxFound": true,
    "nullMx": false,
    "disposable": false,
    "roleBased": false
  },
  "nextAction": "send_verification_link"
}
```

Practical build notes

- Use `validator.js` or a similarly mature library for server-side validation rather than hand-written regexes. [S7]
- In Node.js, `dns.promises.resolveMx()` is the natural MX lookup primitive, and the Node docs show the returned object shape with `priority` and `exchange`. [S6]
- Respect DNS uncertainty. Cache results, but do not assume permanent truth from one lookup; mail routing can change.
- Treat disposable-domain checks as configurable policy, not immutable protocol truth. False positives are possible, and allowlists matter. [S8]

- Protect the endpoint against abuse. Email-checker APIs are attractive for account enumeration and dictionary attacks.

Suggested build roadmap

Build vs buy

If the need is straightforward signup validation, a self-built DNS-plus-verification stack is often enough. If the product needs high-volume sales enrichment, bounce forecasting, mailbox scoring, or historical provider reputation data, a third-party verifier can be justified. That is not because the RFC basics are too hard, but because some of the highest-value signals are proprietary and operational rather than purely protocol-based.

Option	Best when	Upside	Tradeoff
Self-built MVP	You need core validation only	Low cost, fully controllable, easy to ship	You own DNS quirks, sender setup, and ongoing
Self-built + async probe	You need extra signal but not a vendor dependency	Better ranking and risk scoring	Higher complexity and false-confidence risk
Third-party verification API	You need speed, vendor data, or enterprise workflows	Fastest path to broad coverage	Recurring cost, vendor lock-in, opaque heuristics
Hybrid	You want control on core checks and selective enrichment	Balanced cost and capability	Requires thoughtful orchestration

Final recommendation

Build the checker in this order: syntax, DNS route validation, disposable-domain policy, then verification email flow. Add SMTP probing only as an asynchronous enrichment layer. This delivers the best accuracy-to-complexity ratio for most products.

Source Notes

All links below were reviewed during report preparation on April 2, 2026. Official standards and vendor documentation can change, so regenerating this report later is the right way to refresh the conclusions.

S1. RFC 5322: Internet Message Format
https://datatracker.ietf.org/doc/html/rfc5322
Defines the core addr-spec grammar: local-part@domain.

S2. RFC 5321: Simple Mail Transfer Protocol
https://datatracker.ietf.org/doc/rfc5321/
Defines SMTP behavior, local-part handling, implicit MX fallback, and recipient-verification caveats.

S3. RFC 7505: A "Null MX" No Service Resource Record for Domains That Accept No Mail
https://datatracker.ietf.org/doc/html/rfc7505
Defines the MX 0 . pattern that explicitly signals a domain does not accept mail.

S4. RFC 6531: SMTP Extension for Internationalized Email
https://datatracker.ietf.org/doc/rfc6531/
Defines SMTPUTF8 support for internationalized mailbox addresses.

S5. MDN: <input type="email">
https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements/input/email
Documents browser-level email formatting validation.

S6. Node.js DNS API
https://nodejs.org/api/dns.html
Documents resolveMx and related DNS lookups used in Node implementations.

S7. validator.js
https://github.com/validatorjs/validator.js
Documents isEmail and normalizeEmail, including provider-specific normalization options and RFC caveats.

S8. disposable-email-domains
https://github.com/disposable-email-domains/disposable-email-domains
Maintained public disposable-domain list used by production systems such as PyPI.

S9. Google Workspace Admin Help: Email sender guidelines
https://support.google.com/a/answer/81126
Current Gmail delivery requirements, including authentication and DNS expectations.

Original visuals in this PDF were generated locally for this report and are based on the cited standards and documentation rather than copied screenshots.

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